

EduTech Solutions

Internship Completion Report

Intern Details

Intern Name: Disha Mourya

Internship Start Date: June 4, 2026

Internship Duration: 2 Months

Intern ID: EDU-2026-437

Internship Domain: Data Analytics Intern

1. Project Title

E-Commerce Sales Dashboard using Tableau

2. Project Objective

The primary objective of this project is to develop an interactive and insightful E-Commerce Sales Dashboard using Tableau by analyzing the Brazilian Olist E-Commerce dataset. The dashboard is designed to transform raw transactional data into meaningful business insights that support informed decision-making.

The project focuses on understanding sales performance, customer purchasing behavior, payment methods, product categories, and order status. By integrating multiple datasets into a unified dashboard, businesses can efficiently monitor their overall performance through key performance indicators (KPIs) and interactive visualizations.

Another important objective is to enhance practical knowledge of data analytics by applying concepts such as data cleaning, relationship modeling, KPI creation, dashboard design, and data visualization. Throughout this project, Tableau was used to convert complex datasets into a simple, user-friendly dashboard that helps stakeholders identify business trends and opportunities.

This project also demonstrates the importance of business intelligence tools in solving real-world business problems through data-driven analysis and visualization.

3. Project Description in Detail

The **E-Commerce Sales Dashboard** is a Business Intelligence project developed using Tableau Desktop. The project utilizes the **Olist Brazilian E-Commerce Dataset**, which contains real-world transactional data including customer details, product information, orders, payments, and delivery status.

The development process began with understanding the dataset structure and identifying the relationships between different tables. Several CSV files were used, including customer information, order details, product data, payment information, and product category translations. Before creating the dashboard, each dataset was carefully examined to identify missing values, duplicate records, and data inconsistencies. Data cleaning ensured that the final analysis was accurate and reliable.

After preprocessing, relationships were established between the datasets using common keys such as **Customer ID**, **Order ID**, **Product ID**, and **Product Category Name**. This relational model enabled data from multiple tables to be analyzed together within Tableau.

The dashboard includes four important Key Performance Indicators (KPIs):

- **Total Orders**
- **Total Customers**
- **Total Revenue**
- **Average Order Value**

These KPIs provide a quick summary of the overall business performance and help decision-makers understand the current state of sales activities.

To provide deeper insights, several visualizations were created. The **Monthly Sales Trend** line chart shows how sales changed over time, helping identify seasonal patterns and business growth. The **Top 10 Product Categories** chart highlights the highest-performing product categories based on revenue. The **Order Status Distribution** chart displays the number of orders across different order statuses such as Delivered, Shipped, Processing, and Cancelled. Additionally, the **Payment Type Distribution** chart illustrates the contribution of different payment methods, including Credit Card, Debit Card, Voucher, and Boleto.

The dashboard was designed with a clean and professional layout, making it easy for users to interpret business performance at a glance. Appropriate formatting, consistent color schemes, and clear labels were applied to improve readability and user experience.

Overall, this project demonstrates practical knowledge of **data cleaning, data modeling, dashboard development, KPI analysis, and business intelligence reporting** using Tableau. It

provides valuable insights that can help organizations monitor sales performance, understand customer behavior, optimize product strategies, and support data-driven business decisions.

4. Timeline Overview

Task	Activities Planned	Activities Completed
Task 1	Dataset Collection	Collected the Olist Brazilian E-Commerce dataset and identified all required CSV files for analysis.
Task 2	Data Cleaning	Checked all datasets for missing values, duplicate records, and formatting issues. Cleaned the data to ensure consistency.
Task 3	Data Relationship	Established relationships between Orders, Customers, Products, Payments, Order Items, and Product Category tables in Tableau.
Task 4	KPI Development	Created business KPIs including Total Orders, Total Customers, Total Revenue, and Average Order Value.
Task 5	Dashboard Development	Designed interactive visualizations including Monthly Sales Trend, Top Product Categories, Order Status Distribution, and Payment Type Distribution.
Task 6	Dashboard Formatting	Improved dashboard layout, alignment, colors, fonts, labels, and overall presentation
Task 7	Dashboard Testing	Verified dashboard calculations, relationships, filters, and visualization accuracy
Task 8	Documentation	Prepared README.md, Project Report, Dashboard screenshots, and GitHub repository for project submission.

5. Project Development Methodology

The E-Commerce Sales Dashboard project followed a structured Business Intelligence workflow to ensure accurate analysis and effective visualization.

Requirement Analysis

The project started by understanding the business problem and identifying the key metrics required for an E-Commerce dashboard. The primary goal was to analyze sales performance, customer activity, product categories, payment methods, and order status.

Data Collection

The Olist Brazilian E-Commerce dataset was collected, containing multiple related datasets such as Orders, Customers, Products, Payments, Order Items, and Product Category Translation.

Data Cleaning

Each dataset was carefully analyzed to identify blank values, duplicate records, and inconsistent formatting. Data quality checks were performed to improve reliability before analysis.

Data Modeling

Relationships were created between different tables using common fields like Order ID, Customer ID, Product ID, and Product Category Name. This relational model allowed Tableau to combine data efficiently.

Dashboard Development

Business KPIs and visualizations were developed using Tableau Desktop. Charts were formatted using proper colors, labels, titles, and layout to create a clean and professional dashboard.

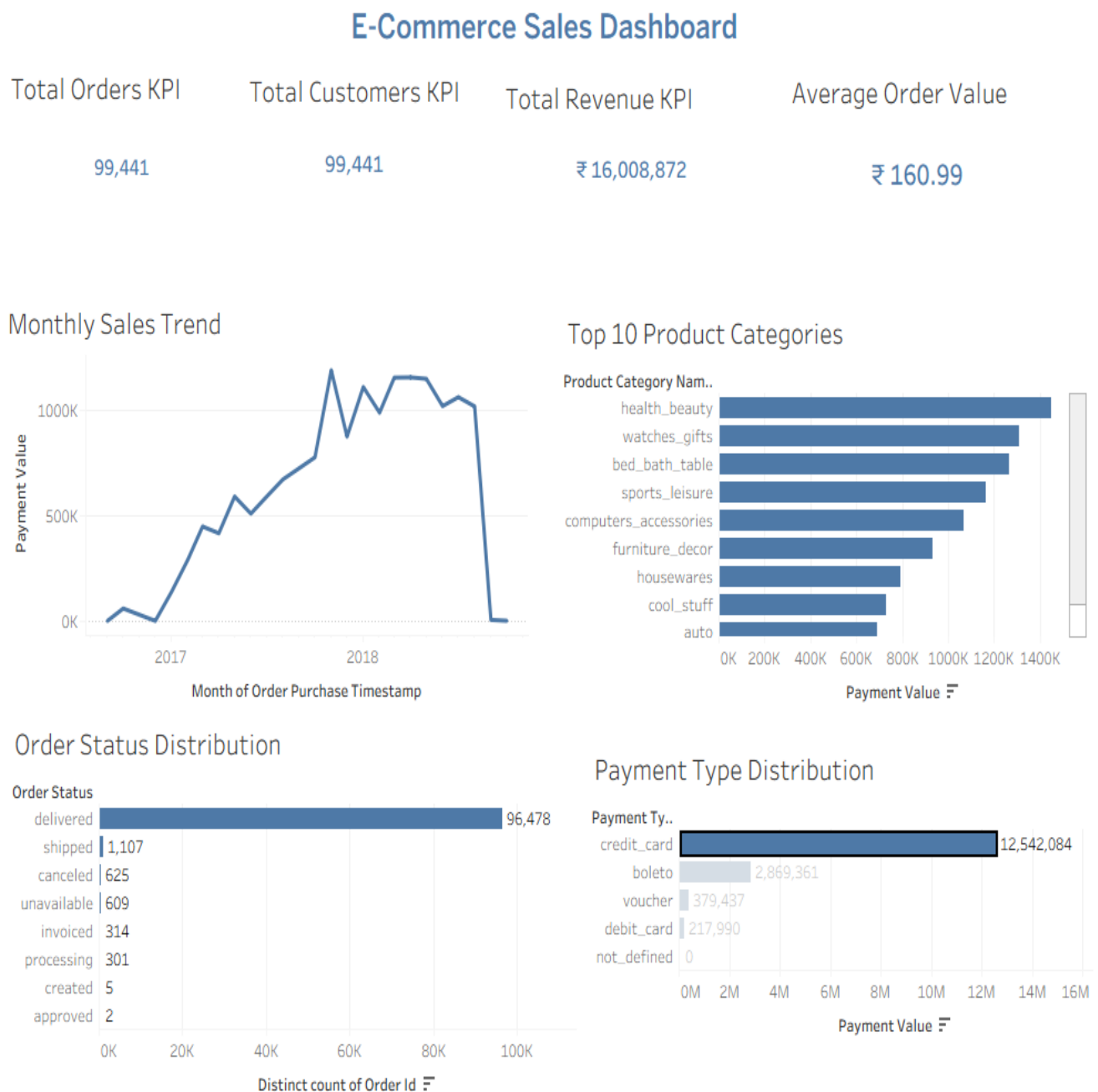
Testing & Validation

All calculations, KPIs, relationships, and visualizations were verified to ensure the dashboard produced accurate business insights. Different scenarios were tested to validate dashboard performance.

6. Output Screenshots

The following screenshots should be included in the final project report.

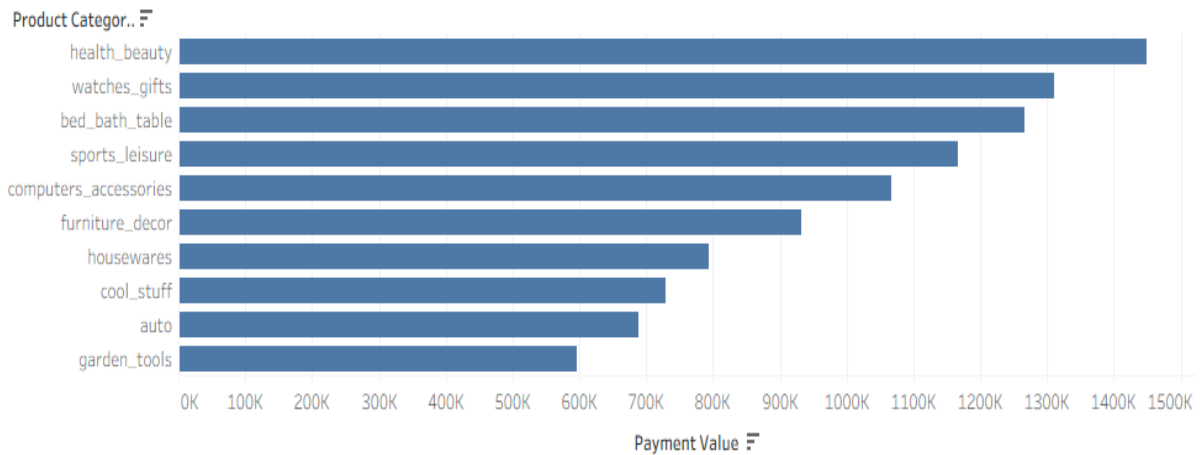
Final Dashboard



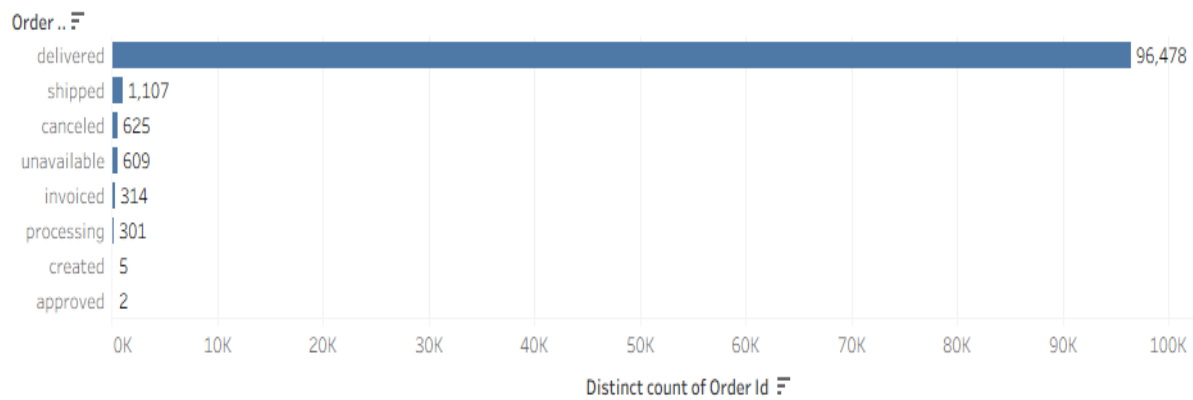
Monthly Sales Trend



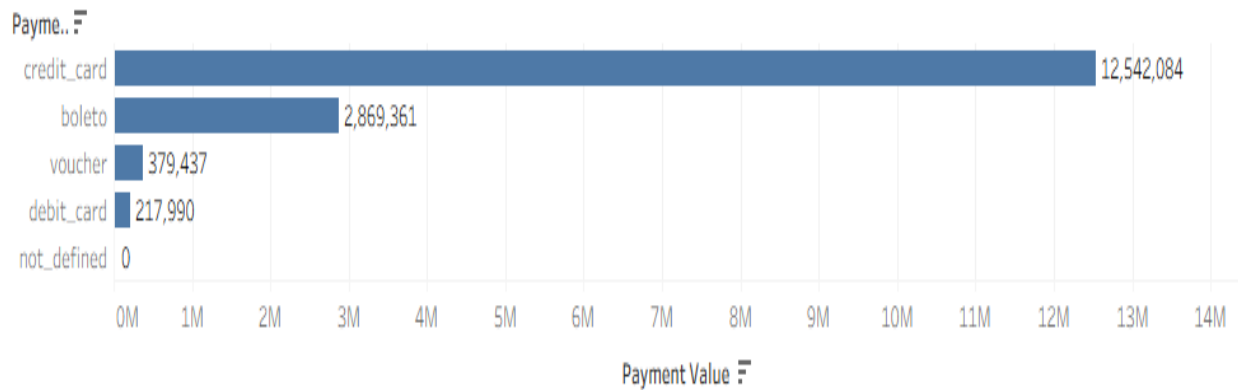
Top 10 Product Categories



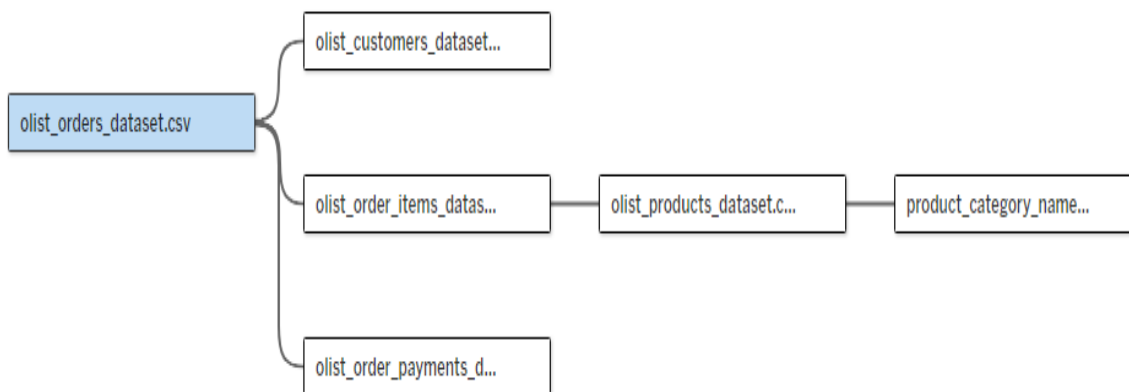
Order Status Distribution



Payment Type Distribution



Relationship Model (Data Source Page)



7. Challenges Faced

Challenge 1: Managing Multiple Datasets

The project required working with multiple CSV files containing customers, orders, products, payments, and order items. Understanding the relationship between these datasets was initially challenging.

Resolution:

Relationships were created carefully using primary fields such as Customer ID, Order ID, Product ID, and Product Category Name. After validating the relationships, data from different tables could be analyzed together accurately.

Challenge 2: Handling Missing Values

Some datasets contained blank values, particularly in delivery-related date fields such as carrier delivery date and customer delivery date.

Resolution:

Each dataset was analyzed individually to identify missing values. Only relevant fields were used for dashboard analysis while maintaining data consistency.

Challenge 3: Dashboard Layout Design

Creating a professional dashboard layout was difficult because multiple KPI cards and charts needed to fit properly within the dashboard area.

Resolution:

The dashboard was redesigned several times by adjusting chart sizes, using floating objects where appropriate, aligning KPI cards, and applying consistent formatting to achieve a clean and organized layout.

Challenge 4: Tableau Configuration

During development, there were issues related to Tableau setup, data connections, and dashboard formatting, which affected the initial workflow.

Resolution:

The configuration issues were resolved by correctly connecting datasets, validating relationships, and testing each visualization individually before integrating them into the final dashboard.

8. Learning's & Skills Acquired

Throughout this internship project, I gained valuable practical experience in Business Intelligence, Data Analytics, and Dashboard Development. Working with a real-world e-commerce dataset helped me understand the complete workflow of a data analytics project, from raw data preparation to final dashboard creation.

The project strengthened my analytical thinking and improved my ability to identify business insights from complex datasets. It also enhanced my confidence in using Tableau for creating professional dashboards and presenting data in a meaningful way.

Technical Skills

During the project, I learned and improved the following technical skills:

- Tableau Dashboard Development
- Data Cleaning and Preprocessing
- Data Relationship Modeling

- KPI Design and Business Metrics
- Data Visualization Techniques
- Microsoft Excel
- CSV Data Handling
- Business Intelligence Concepts
- Interactive Dashboard Design
- GitHub Repository Management
- Documentation and Report Preparation

Soft Skills

The internship also helped me develop several professional skills, including:

- Analytical Thinking
- Problem Solving
- Time Management
- Self-Learning
- Research Skills
- Communication Skills
- Attention to Detail
- Documentation Skills
- Presentation Skills
- Project Planning

9. Testimonial from Team

- The E-Commerce Sales Dashboard project provided valuable hands-on experience in analyzing business data and developing interactive dashboards using Tableau.
 - Throughout the internship, I learned how to work with multiple datasets, establish relationships between tables, clean and prepare data, create meaningful KPIs, and design professional dashboards that support business decision-making.
 - Working on this project improved my understanding of Business Intelligence concepts and strengthened my practical knowledge of Tableau, Excel, and data visualization techniques. Continuous learning, problem-solving, and regular project reviews helped me complete the dashboard successfully and enhanced both my technical and professional skills.
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10. Conclusion

The **E-Commerce Sales Dashboard** was successfully developed using Tableau by integrating multiple datasets from the Olist Brazilian E-Commerce database.

The project demonstrated the complete Business Intelligence workflow, including:

- Data Collection
- Data Cleaning
- Relationship Modeling
- KPI Development
- Dashboard Design
- Data Visualization
- Documentation

The dashboard provides meaningful insights into sales performance, customer activity, payment methods, product categories, and order status. These insights help businesses monitor their operations, identify trends, and make informed decisions based on reliable data.

Overall, this internship project enhanced my practical knowledge of Tableau and Business Intelligence while improving my analytical thinking and problem-solving abilities. It also gave me real-world experience in developing professional dashboards that can support organizational decision-making.

11. Future Scope

The dashboard can be further improved by adding more advanced features, such as:

- Region-wise Sales Analysis
- Customer Segmentation
- Profit Analysis
- Product Return Analysis
- Sales Forecasting
- Customer Lifetime Value (CLV)
- Interactive Dashboard Filters
- Drill-down Analysis
- Mobile Dashboard Optimization
- Integration with Live Data Sources

These enhancements would make the dashboard more powerful and suitable for real-time business monitoring.

12. Acknowledgements

I would like to express my sincere gratitude to my mentor and the entire **EduTech Solutions** team for their continuous guidance, support, and encouragement throughout my internship.

I am thankful for the opportunity to work on a real-world Data Analytics project that allowed me to apply theoretical concepts to practical business scenarios. This internship significantly improved my understanding of Tableau, Business Intelligence, Data Visualization, and Dashboard Development.

I also extend my gratitude to my faculty members and everyone who supported me during this internship. Their motivation and valuable suggestions helped me successfully complete this project.

Finally, I would like to thank the creators of the **Olist Brazilian E-Commerce Dataset**, Tableau, Microsoft Excel, GitHub, and the open-source community for providing the resources that made this project possible.